Artwork Code : XXXXXXXXXX-XX	Artwork Req. No: -
Quality of Paper/Board : Super Sun Shine	Size of Artwork (In mm) : 210x133
Quality of Gum :	GSM of Paper/Board : 60 GSM
Colour Code : BLACK Pantone Pantone Pantone Pantone Pantone Pantone Pantone	
Barcode Information:	
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Packing: 500ml PLB	Plant Location : MF 02
Country: Malaysia	Language : English
Ref. Code Creation/Blockage Note:	Artwork Control Key No.:
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FLUSODEX

SODIUM CHLORIDE & GLUCOSE INTRAVENOUS INFUSION BP (0.45% w/v & 5% w/v)

For Intravenous use only.

1. COMPOSITION

Active Ingredient:

Each 100 ml contains:

Glucose Anhydrous BP	5.00 g
Sodium Chloride BP	0.45 g

Excipients:

Water for Injections BP

2. PRODUCT DESCRIPTION

Flusodex is available as 500 mL Nipple head Plastic Bottle & Eurohead Plastic Bottle.

It is a clear, colourless or faintly straw coloured solution essentially free from visible particles.

3. INDICATION

Dehydration

Light Sodium and Chloride depletion

Hypochloremic alkalosis

Energy supply

Vehicle solution for supplementary medication.

4. DOSAGE AND ADMINISTRATION

Dose varies depending on the clinical condition and size of the patient.

Approximately 1000ml/day

Drop rate: Up to 120 drops/min corresponding to 360-540 ml/h.

Route of administration I.V.

5. CONTRAINDICATIONS

The solution is contraindicated in patients presenting with:

- Known hypersensitivity to the product
- Extracellular hyperhydration or hypervolaemia
- Fluid and sodium retention
- Severe renal insufficiency (with oliguria/anuria)
- Uncompensated cardiac failure
- Hypernatraemia or hyperchloraemia
- General oedema and ascitic cirrhosis

Clinically significant hyperglycaemia. The solution is also contraindicated in case of uncompensated diabetes, other known glucose intolerances (such as metabolic stress situations), hyperosmolar coma or hyperlactataemia.

6. WARNING AND PRECAUTION

Excessive administration of Sodium Chloride causes hypernatraemia, the most serious effect of which is dehydration of internal organs, especially the brain. Excess chloride in the body may cause a loss of bicarbonate with an acidifying effect.

7. PREGNANT AND BREAST FEEDING WOMAN

Data from preclinical and clinical studies of the use of this product in these conditions are not available. Therefore Sodium Chloride & Glucose Intravenous Infusion BP (0.45% w/v & 5% w/v) should be administered with caution during pregnancy and lactation.

It has been suggested that if used during labour, the glucose load on the mother may lead to foetal hyperglycaemia, hyperinsulinaemia, and acidosis, with subsequent neonatal hypoglycaemia and jaundice. Others have found no evidence of such an effect.

Sodium Chloride & Glucose Intravenous Infusion BP (0.45% w/v & 5% w/v) should be administrated with special caution for pregnant women during labour particularly if administered in combination with oxytocin due to the risk of hyponatraemia.

8. EFFECTS ON DRIVING AND MACHINE OPERATION

There is no information on the effects of Sodium Chloride & Glucose Intravenous Infusion BP (0.45% w/v & 5% w/v) on the ability to operate an automobile or other heavy machinery.

9. INTERACTION, COMPATIBILITY, INCOMPATIBILITY

No studies have been conducted.

Both the glycaemic and effects on water and electrolyte balance should be taken into account when administering Sodium Chloride & Glucose Intravenous Infusion BP (0.45% w/v & 5% w/v) to patients treated with other substances that affect glycaemic control, or fluid and/or electrolyte balance.

Caution is advised in patients treated with

- lithium. Renal sodium and lithium clearance may be increased during administration and can result in decreased lithium levels.
- corticosteroids, which are associated with the retention of sodium and water (with oedema and hypertension).
- diuretics, beta-2 agonist, or insulin, whom increase the risk of hypokalemia

10. SIDE EFFECT

System Organ Class	Adverse reactions (Preferred terms)	Frequency
Immune system disorders	anaphylactic reaction, * hypersensitivity*	Not known
Metabolism and nutrition disorders	hypematraemia, hyperglycaemia,	Not known
Vascular disorders	phlebitis	Not known
Skin and subcutaneous tissue disorders	rash, pruritus	Not known
General disorders and administration site conditions	injection site reactions including: pyrexia chills infusion site pain infusion site vesicles	Not known

**Potential manifestation in patients with allergy to corn*

Other adverse reactions reported with isotonic saline and glucose infusions include:

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- Hyponatraemia, which may be symptomatic
- Hyperchloraemic acidosis

Adverse reactions may be associated to the medicinal product(s) added to the solution; the nature of the additive will determine the likelihood of any other adverse reactions.

11. OVERDOSE: SYMPTOM AND TREATMENT

Excess administration of Sodium Chloride & Glucose Intravenous Infusion BP (0.45% w/v & 5% w/v) solution can cause:

- Hyperglycaemia, adverse effects on water and electrolyte balance and corresponding complications. For example, severe hyperglycaemia and severe dilutional hyponatraemia and their complications, can be fatal.
- Hyponatraemia (which can lead to CNS manifestations, including seizures, coma, cerebral oedema and death).
- Hypernatraemia especially in patients with renal impairment.
- Fluid overload (which can lead to central and/or peripheral oedema).

A clinically significant overdose of Sodium Chloride & Glucose Intravenous Infusion BP (0.45% w/v & 5% w/v) solution may, therefore, constitute a medical emergency.

When assessing an overdose, any additives in the solution must also be considered. Interventions include discontinuation of Sodium Chloride & Glucose Intravenous Infusion BP (0.45% w/v & 5% w/v) solution administration, dose reduction, administration of insulin and other measures as indicated for the specific clinical constellation.

12. PHARMACODYNAMIC PROPERTIES (INCLUDING ATC):

Pharmacotherapeutic group "Electrolytes with Carbohydrates",

ATC code: "B05BB02".

Sodium Chloride & Glucose Intravenous Infusion BP (0.45% w/v & 5% w/v) is an hypotonic and hyperosmolar solution of sodium chloride and glucose.

The pharmacodynamic properties of this solution are those of its components (glucose, sodium and chloride).

Ions, such as sodium, circulate through the cell membrane, using various mechanisms of transport, among which is the sodium pump (Na^+/K^+-ATPase). Sodium plays an important role in neurotransmission and cardiac electrophysiology, and also in renal metabolism.

Chloride is mainly an extracellular anion. Intracellular chloride is in high concentration in red blood cells and gastric mucosa. Reabsorption of chloride follows reabsorption of sodium.

Glucose is the principal source of energy in cellular metabolism. The glucose in this solution provides a caloric intake of 200kcal/l.

13. PHARMACOKINETIC PROPERTIES

The pharmacokinetic properties of this solution are those of its components (glucose, sodium and chloride).

After injection of radiosodium (^{24}Na), the halflife is 11 to 13 days for 99% of the injected Na and one year for the remaining 1%. The distribution varies according to tissues: it is fast in muscles, liver, kidney, cartilage and skin; it is slow in erythrocytes and neurones; it is very slow in the bone. Sodium is predominantly excreted by the kidneys, but (as described earlier) there is extensive renal reabsorption. Small amounts of sodium are lost in the faeces and sweat.

The two main metabolic pathways of glucose are gluconeogenesis (energy storage) and glycogenolysis (energy release). Glucose metabolism is regulated by insulin.

14. INSTRUCTION FOR USE

Method of administration

The administration is performed by intravenous infusion.

Sodium Chloride & Glucose Intravenous Infusion BP (0.45% w/v & 5% w/v) is hypotonic and hyperosmolar, due to the glucose content. It has an approximate osmolarity of 432mOsm/l.

Precautions to be taken before manipulating or administering the product

Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration.

Do not administer unless the solution is clear and the seal is intact. Administer immediately following the insertion of infusion set.

The solution should be administered with sterile equipment using an aseptic technique.

The equipment should be primed with the solution in order to prevent air entering the system.

Do not use plastic containers in series connections. Such use could result in air embolism due to residual air being drawn from the primary container before the administration of the fluid from the secondary container is completed.

Pressurizing intravenous solutions contained in flexible plastic containers to increase flow rates can result in air embolism if the residual air in the container is not fully evacuated prior to administration. Use of a vented intravenous administration set with the vent in the open position could result in air embolism. Vented intravenous administration sets with the vent in the open position should not be used with flexible plastic containers.

Do not remove unit from overwrap until ready for use. The inner bag maintains the sterility of the product.

Monitoring:

Fluid balance and the concentrations of glucose and electrolytes (especially sodium) in plasma must be monitored during administration. Electrolyte supplementation may be indicated according to the clinical needs of the patient.

15. PACKAGING SIZE

500 mL in Nipple head Plastic Bottle & Eurohead Plastic Bottle.

16. STORAGE CONDITION

Store at a temperature not exceeding 30°C.

Date of revision: 10-Oct-2024

MAL No.:-

17. MANUFACTURER

Otsuka

Otsuka Pharmaceutical India Private Limited

Survey No.199 to 201, 208 To 210,
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18. Product Registration Holder, Importer & Distributor

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Pusat Dagangan, PJCC Jalan PJS
5/28 46150 Petaling Jaya, Selangor, Malaysia

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